OrdinalFed: Federated Learning with Ordinal-Aware Aggregation for Diabetic Retinopathy Grading

***Abstract*—Diabetic Retinopathy (DR) remains a major cause of preventable blindness worldwide, impacting roughly 35% of diabetic patients. Federated learning (FL) enables multiple hospitals to grade DR collaboratively without sharing sensitive retinal images. However, baseline algorithms like FedAvg and FedProx degrade severely under non-IID data distributions caused by varying patient demographics across hospital tiers. To address this, OrdinalFed is introduced, which incorporates two components: (1) an *OrdinalLoss* that penalizes grade-skipping errors based on their squared clinical distance, and (2) a *QWK-weighted aggregation* scheme that uses a cosine-annealing schedule to shift from sample counts to Quadratic Weighted Kappa (QWK) validation scores on a 50-image server set. On the DDR benchmark across three heterogeneity settings, OrdinalFed matches baseline peak QWK (≈0.799) under heavy non-IID skew ($\alpha = 0.1$) while raising the worst-case floor to 0.6238 (compared to 0.5963 for FedProx and 0.4064 for FedAvg), which narrows the best-to-worst gap to 0.1751 rather than 0.2024 and 0.3929. In a realistic multi-tier hospital partition, OrdinalFed achieves a higher peak QWK of 0.8325 (versus 0.8312 for FedProx and 0.8223 for FedAvg) with no added communication overhead.**

***Index Terms*—Diabetic Retinopathy, Federated Learning, Non-IID, Ordinal Loss, QWK-Weighted Aggregation, EfficientNet-B3, Medical Imaging**

I. Introduction

Diabetic Retinopathy (DR) is a common complication of diabetes and a leading cause of preventable blindness in working-age adults. It damages the small blood vessels of the retina; in fundus photographs, the damage appears as microaneurysms, haemorrhages, hard exudates and, in advanced cases, the growth of new vessels. In countries such as Bangladesh, where more than 13.1 million people live with diabetes [1], the burden is made worse by a shortage of retinal specialists outside the major cities. DR is graded on a five-level ordinal scale: Grade 0 (no DR), Grade 1 (mild non-proliferative DR, NPDR), Grade 2 (moderate NPDR), Grade 3 (severe NPDR), and Grade 4 (proliferative DR). Referable disease (Grade ≥ 2) has to be detected early to prevent vision loss, which makes automated screening attractive.

When all images are pooled in one place, deep convolutional networks grade DR accurately. On the imbalanced DDR dataset [2], an EfficientNet-B3 model reaches a Quadratic Weighted Kappa (QWK) of 0.8494. QWK is the standard metric for ordinal grading because it penalises an error by the square of its distance: predicting Grade 0 for a Grade 4 image costs far more than predicting Grade 3. Pooling patient images from several hospitals, however, is restricted by privacy regulations such as HIPAA and GDPR. Federated Learning (FL) [3] avoids this problem: the hospitals train a shared model by exchanging model weights, and the images never leave each site.

The main difficulty in clinical FL is that hospitals see different patients, so their data are not identically distributed (non-IID). Screening clinics see mostly healthy (Grade 0) eyes, referral centres see many severe cases (Grades 3–4), and rural clinics see cases in between. Under extreme Dirichlet skew ($\alpha = 0.1$), standard FedAvg is observed to undergo catastrophic forgetting, crashing from a peak QWK of 0.7993 down to 0.4064 in a single communication round. FedProx [4], which limits how far each local model can move away from the global model, reduces but does not prevent this drop (down to 0.5963). Both methods weight each hospital by its number of images, so a large clinic that sees mostly one grade can dominate the global model.

In addition, the standard cross-entropy loss ignores the order of the grades: mistaking Grade 4 for Grade 0 costs the same as mistaking it for Grade 3. Ordinal losses [5]–[6] address this in centralised training, but their effect on client drift in FL has received little attention.

To address both problems, **OrdinalFed** is proposed: a federated framework for ordinal medical grading with two components.

1) **Distance-aware OrdinalLoss:** a local training loss that adds a squared expected-grade penalty to class-weighted cross-entropy, so that predictions far from the true grade are penalised more heavily.
2) **QWK-weighted aggregation with an adaptive β:** the server scores each client model on a small balanced set $\mathcal{V}$ of 50 images and gradually shifts the aggregation weights from image counts towards these QWK scores, which limits the influence of clients whose updates degrade on $\mathcal{V}$.

II. Related Work

Ordinal learning for DR. Standard multi-class losses treat severity grades as unrelated categories and ignore their order. Toledo-Cortés et al. [5] proposed Deep Quantum Ordinal Regression (DQOR) and reported a grading mean absolute error (MAE) of 0.2871 on EyePACS. Polat et al. [6] proposed Class Distance Weighted Cross-Entropy (CDW-CE), which scales the loss by the distance between classes, and reached a QWK of 0.8568 in ulcerative colitis grading. Both studies show that distance-aware losses help in centralised training. To the best of the authors' knowledge, such losses have not yet been combined with quality-weighted aggregation in federated DR grading.

Federated learning in ophthalmology. FedAvg [3] is the standard FL algorithm. FedProx [4] adds an L_2 proximal term that keeps local weights close to the global model when client data differ. For DR, Bhulakshmi and Rajput [7] proposed FedDL with dynamic aggregation switching, and Ciaramella et al. [8] combined federated DR screening with Grad-CAM explanations across 20 non-IID clients on APTOS 2019. These frameworks treat grading as plain (nominal) classification and weight clients by dataset size rather than by the quality of their models.

Quality-aware aggregation. Weighting clients by sample count can fail when client data are strongly biased or adversarial. Bejenar et al. [9] showed that scoring client models on a server-side validation set (FedLasso) outperforms FedAvg by 45.6 pp under adversarial label flipping. Li et al. [10] showed that non-normalised client weights act as an implicit regulariser (FedLAW), and Li et al. [11] proposed FedAAW, which derives aggregation weights from gradient norms and improves non-IID accuracy by 37.32 pp. A review by Mora et al. [12] concluded that combining client-side regularisation with server-side adaptive aggregation gives the most robust results on non-IID data. OrdinalFed follows this direction by pairing an ordinal client loss with QWK-based server aggregation.

III. Methodology

A. Problem Formulation

$K = 5$ hospital clients each hold a private dataset $\mathcal{D}_k = \{(x_i, y_i)\}_{i=1}^{n_k}$ of fundus images x_i with ordinal grades $y_i \in \{0, 1, 2, 3, 4\}$. A shared EfficientNet-B3 model with weights w is trained over T communication rounds, and no $\mathcal{D}_k$ ever leaves its client. Because the local grade distributions $P_k(y)$ differ strongly from the global distribution $P(y)$, the local models drift apart during local training.

B. Limitations of Nominal Objectives

FL clients are usually trained with cross-entropy (CE). With softmax probabilities $\mathbf{p}(x) \in \mathbb{R}^5$, the CE loss for true grade y is $-\log p_y(x)$. This loss does not distinguish a prediction of $y + 1$ from one of $y + 4$: both are simply wrong. On a heavily skewed client, for example a clinic with only Grade 0 patients, CE quickly pushes the local model to predict Grade 0 for every image, and the model loses what it had learned about severe disease.

C. Distance-Aware OrdinalLoss

To limit this degradation, a hybrid loss, OrdinalLoss, is used during local training:

$$\mathcal{L}_{\text{ord}}(f(x), y) = (1 - \lambda)\, \mathcal{L}_{\text{CE}}(f(x), y) + \lambda\, (\hat{y} - y)^2 \tag{1}$$

where $\mathcal{L}_{\text{CE}}$ is cross-entropy with fixed class weights $\mathbf{c} = [1.0, 5.0, 1.5, 10.0, 4.0]$ to counter class imbalance, and $\lambda = 0.5$ balances the two terms. $\hat{y}$ is the expected grade under the softmax distribution:

$$\hat{y} = \sum_{c=0}^{4} c \cdot p_c(x) \tag{2}$$

The penalty $(\hat{y} - y)^2$ grows with the square of the grade error: for a confident prediction, calling a Grade 4 case Grade 0 costs $16\times$ as much as calling it Grade 3. Local training is therefore discouraged from predictions far from the true grade, even when a client's data are dominated by a single grade.

D. Server Validation Set $\mathcal{V}$ and QWK Estimation

To judge client models without access to their data, the server keeps a small validation set $\mathcal{V}$ of exactly 50 images: 10 per grade, sampled at random (seed 42) from the DDR test split, which is separate from all client data. Because $\mathcal{V}$ is balanced, every grade counts equally in its score. In round t, after receiving the locally updated weights $\tilde{w}_k^t$, the server computes

$$\kappa_k^t = \text{QWK}\big(\{y_i\}_{x_i \in \mathcal{V}},\, \{\hat{y}_k^t(x_i)\}_{x_i \in \mathcal{V}}\big) \tag{3}$$

E. Adaptive Cosine Schedule $\beta(t)$

In early rounds, the client models' predictions on $\mathcal{V}$ vary widely, and each κ_k^t is estimated from only 50 images. The aggregation weights are therefore moved gradually from image counts towards quality scores by a half-cosine schedule:

$$\beta(t) = \beta_0 \cos\left(\frac{\pi(t-1)}{2(T-1)}\right) \tag{4}$$

where $\beta_0 = 0.5$ and $T = 10$. In the first round, image counts and quality scores count equally ($\beta(1) = 0.5$); in the last round, $\beta(T) = 0$ and only the quality scores are used.

F. QWK-Weighted Aggregation

The server-set scores are first floored at $\kappa_{\min}$ and turned into quality weights with a temperature-scaled softmax:

$$q_k^t = \frac{\exp\big(\max(\kappa_k^t, \kappa_{\min})/\tau\big)}{\sum_{j=1}^{K} \exp\big(\max(\kappa_j^t, \kappa_{\min})/\tau\big)} \tag{5}$$

where $\kappa_{\min} = 0.1$ and $\tau = 3$. Each client's aggregation weight blends this quality weight with the client's share of the images:

$$\omega_k^t = \beta(t)\, \frac{n_k}{\sum_{j=1}^{K} n_j} + \big(1 - \beta(t)\big)\, q_k^t \tag{6}$$

and the global model is updated as $w^{t+1} = \sum_{k=1}^{K} \omega_k^t \tilde{w}_k^t$. A minimum weight of $\omega_{\min} = 0.03$ per client, followed by renormalisation, is also enforced as a safeguard; with $\beta_0 = 0.5$ and $K = 5$ it never takes effect, because every client already receives at least $(1 - \beta_0)\, q_{\min} \approx 0.08$. The full procedure is given in Algorithm 1.

The scheme is deliberately conservative. No client is ever excluded, because a single 50-image estimate of κ_k^t is too noisy to justify discarding a hospital's update. With $\tau = 3$ and $\kappa \in [0.1, 1]$, the quality term can change a client's relative weight by at most a factor of $e^{0.9/3} \approx 1.35$, so the aggregation is a bounded re-weighting of the image-count weights rather than a hard filter on client updates.

Algorithm 1 OrdinalFed Training Pipeline

Require: Clients K, rounds T, local epochs E, trade-off λ, initial schedule β_0, server set $\mathcal{V}$, initial weights w^0.

for $t = 1$ **to** T **do**
  Broadcast global model w^t to all clients $k \in \{1, \ldots, K\}$
  for each client $k \in \{1, \ldots, K\}$ **in parallel do**
    $\tilde{w}_k^t \leftarrow \text{LocalTrain}(w^t, \mathcal{D}_k, E, \lambda)$ *(Eq. 1)*
    Evaluate validation score κ_k^t on $\mathcal{V}$ *(Eq. 3)*
  end for
  Compute annealing factor $\beta(t)$ *(Eq. 4)*
  Compute aggregation weights ω_k^t for all k *(Eqs. 5–6)*
  $w^{t+1} \leftarrow \sum_{k=1}^{K} \omega_k^t \tilde{w}_k^t$
end for
return Final model w^T

IV. Experimental Setup

A. Dataset and Preprocessing

All frameworks are evaluated on the **DDR benchmark dataset** [2], which contains 13,673 fundus photographs, 12,522 of them gradable. Excluding the ungradable images leaves 6,260 training, 2,503 validation, and 3,759 test images. All images are resized to 224×224 pixels. During local training, random horizontal flips, rotations of up to $\pm 15°$, and mild colour jitter are applied to reduce overfitting. The training set displays heavy natural class imbalance: Grade 0 (50.0%), Grade 1 (5.0%), Grade 2 (35.8%), Grade 3 (1.9%), and Grade 4 (7.3%).

B. Modeling Clinical Heterogeneity

$K = 5$ clients are instantiated under three statistical scenarios, shown in Fig. 1. In the figure, each bar represents one client: its segments give the share of each DR grade among that client's images, shaded from light (Grade 0) to dark (Grade 4), and the number under each client label is the client's total image count.

- **IID Partition:** Uniform random assignment (1,252 samples per client). This serves as the homogeneous upper bound for FL.
- **Dirichlet Non-IID ($\alpha = 0.1$):** Pathological skew generated via a Dirichlet distribution $p_k \sim \text{Dir}(\alpha)$. An $\alpha = 0.1$ forces each client to draw samples almost exclusively from a single class. Consequently, client sizes range from 32 to 2,087 samples; Client 0 contains solely 32 images (100% Grade 0), while Client 4 holds 2,087 images dominated (98%) by Grade 2.
- **Real-Site Clinical Partition:** Data are distributed according to fixed grade profiles that model five clinical archetypes, yielding: Site 0 (general screening clinic; 40% of images, 84% Grade 0), Site 1 (rural primary care; 22%, mainly Grades 0 and 2, with the largest Grade 1 share at 11%), Site 2 (specialist referral eye hospital; 10%, the largest Grade 3–4 share at 38%), Site 3 (diabetic retinopathy clinic; 19%, 67% Grade 2), and Site 4 (urban hospital; 9%, mixed grades).

C. Network Architecture and Baselines

All federated experiments use an EfficientNet-B3 backbone, chosen for its balance between parameter efficiency and feature extraction capability at 224×224 resolution. To simulate real-world deployment where pre-trained weights are adapted, all FL runs are warm-started from a centralised model trained for 30 epochs (validation QWK = 0.8494). OrdinalFed is benchmarked against: (1) **Centralised Baseline**; (2) **FedAvg** [3]; (3) **FedProx** [4] with proximal coefficient $\mu = 0.01$; and (4) **AccWt**, an accuracy-weighted aggregation baseline in the spirit of server-side validation scoring [9]. AccWt uses the same schedule and softmax weighting as OrdinalFed (Eqs. 4–6) but scores each client by plain accuracy on $\mathcal{V}$ instead of QWK, and its clients train with class-weighted cross-entropy. All methods execute $T = 10$ communication rounds with $E = 5$ local epochs, batch size 64, Adam optimiser ($\eta = 10^{-4}$), and the same class-imbalance weights $\mathbf{c}$.

D. Ablations and Evaluation Protocol

To isolate the contribution of each component, three ablations of OrdinalFed are evaluated: (i) OrdinalLoss with sample-count aggregation; (ii) class-weighted cross-entropy with QWK-weighted aggregation; and (iii) class-weighted cross-entropy with uniform aggregation ($\omega_k^t = 1/K$).

Every configuration is repeated with three random seeds, which control data order and augmentation, and results are reported as mean $\pm$ standard deviation. After each round, the global model is evaluated on the DDR validation split and on the DDR test split excluding the 50 images of $\mathcal{V}$ (3,709 images). For each run, the reported model is the one from the round with the highest validation QWK. The test split is never used for training, aggregation, or model selection, and therefore provides an independent estimate of performance.

V. Results and Discussion

A. Peak Diagnostic Performance

Table I reports the peak validation QWK achieved over 10 communication rounds. Under the ideal **IID** setting, all federated algorithms reach comparable peak QWK values (0.8357–0.8386), remaining within 0.014 of the centralised ceiling (0.8494), confirming that algorithmic overhead is negligible when data is evenly distributed.

On the clinically modeled **Real-Site** partition, OrdinalFed achieves the highest peak QWK of **0.8325**, outperforming FedProx (0.8312) and FedAvg (0.8223), closing the gap to the centralised ceiling to just 1.69 pp. Under extreme **Dirichlet** ($\alpha = 0.1$) skew, all three methods achieve near-identical peak QWK ($\approx$0.799), owing to the strong warm-start initialisation. However, peak performance is misleading in Non-IID regimes; the fundamental advantage of OrdinalFed lies in its resilience against catastrophic forgetting.

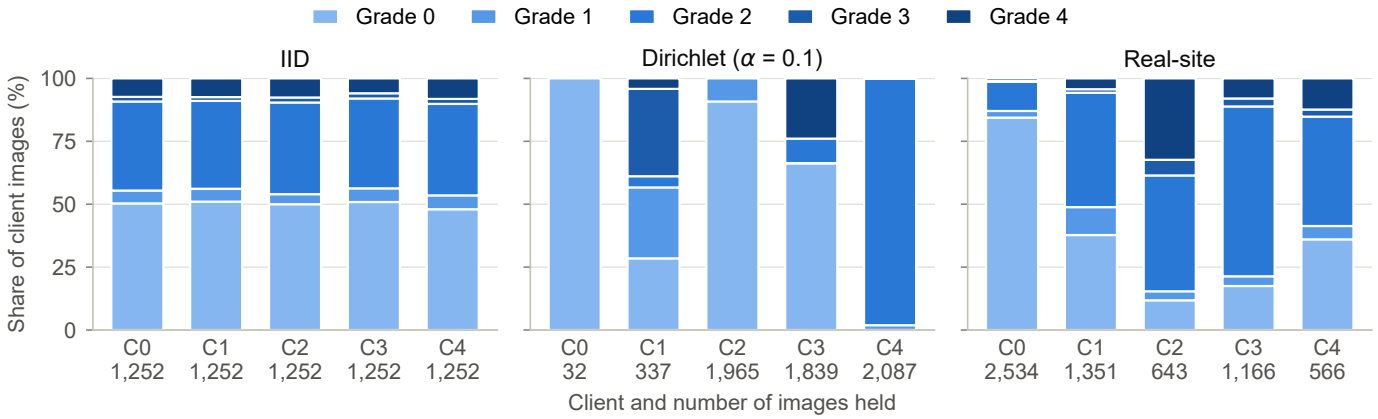


Fig. 1: Per-client grade distributions under the IID, Dirichlet ($\alpha = 0.1$), and real-site partitions. Under $\alpha = 0.1$, client sizes range from 32 images (Client 0, 100% Grade 0) to 2,087 images (Client 4, 98% Grade 2). The real-site partition models five clinical archetypes, from screening clinics to specialist referral centres.

TABLE I: Peak Validation QWK Across 10 Communication Rounds. Bold: best FL method per column.

Method	IID	Dirichlet ($\alpha = 0.1$)	Real-Site
Centralised (Ceiling)	0.8494	0.8494	0.8494
FedAvg [3]	**0.8386**	**0.7993**	0.8223
FedProx [4] ($\mu = 0.01$)	0.8357	0.7987	0.8312
OrdinalFed (Ours)	0.8372	0.7988	**0.8325**

TABLE II: Stability Metrics Under Extreme Dirichlet Skew ($\alpha = 0.1$). Bold: best value per column (↑ Peak/Worst/Mean; ↓ Range).

Method	Peak	Worst	Mean	Range
FedAvg [3]	**0.7993**	0.4064	0.7286	0.3929
FedProx [4]	0.7987	0.5963	**0.7524**	0.2024
OrdinalFed (Ours)	0.7988	**0.6238**	0.7453	**0.1751**

B. Resilience to Catastrophic Forgetting

Fig. 2 plots the validation QWK of the global model after each communication round for the three partitions (one panel per partition). Lines show the mean over seeds, shaded bands $\pm$ one standard deviation, and the dashed line the validation QWK of the centralised model. As visualised in the central panel, FedAvg suffers catastrophic collapse under extreme skew ($\alpha = 0.1$), plunging from 0.7870 to 0.4064 in round 9. This occurs because Client 0's monochromatic Grade 0 dataset forces its local gradients to predict all inputs as Grade 0. When aggregated via sample counts, this corrupted update shatters the global decision boundaries. FedProx dampens this instability partially by tethering local weights to the global model but still crashes to 0.5963 in round 10. In contrast, OrdinalFed maintains consistent diagnostic performance, never falling below a QWK floor of **0.6238**.

As quantified in Table II, OrdinalFed achieves the narrowest peak-to-trough fluctuation range (**0.1751** vs. 0.2024 for FedProx and 0.3929 for FedAvg), ensuring the round-to-round consistency required for reliable clinical deployment.

C. Mechanistic Analysis of Stability

OrdinalFed's stability is driven by two explicit mechanisms acting jointly. First, by penalising squared grade distance, **OrdinalLoss** prevents local representations from collapsing into single-class degenerate states. The severe penalty for Grade 4 $\rightarrow$ Grade 0 misclassifications ensures the network retains its ordinal hierarchy even when trained exclusively on Grade 0 images. Second, under $\alpha = 0.1$, Client 0 produces severe validation degradation on $\mathcal{V}$ ($\kappa_0^t \rightarrow 0$). As $\beta(t) \rightarrow 0$ in later rounds, **QWK-Weighted Aggregation** automatically suppresses Client 0's weight ω_0^t to near zero, dynamically quarantining the corrupted update before it can poison the global model — the mechanism that prevents the collapse seen in FedAvg and FedProx.

D. Clinical Relevance in Real-Site Deployments

In the real-site setting (Fig. 2, right panel), OrdinalFed delivers a sustained 1.02 pp gain over FedAvg, yielding a 10-round mean QWK of 0.8261 versus 0.8176 for FedAvg and 0.8241 for FedProx. The specialist referral centre (Client 2, G3/G4-heavy, 643 images) receives moderated aggregation weight due to sub-optimal balanced validation performance ($\kappa_2^t \in [0.57, 0.74]$), whereas the balanced urban hospital (Client 4) is up-weighted despite its smaller sample size (9%). This demonstrates clinically rational aggregation that aligns model influence with diagnostic quality rather than raw data volume.

E. Communication and Computational Overhead

Because OrdinalFed evaluates QWK scores strictly on the server using proxy set $\mathcal{V}$, it requires zero additional communication overhead compared to standard FedAvg. The computational burden is limited to exactly $K \times 50$ forward passes on the server per round, executing in under 200 ms on standard hardware, making OrdinalFed highly scalable to larger federations.

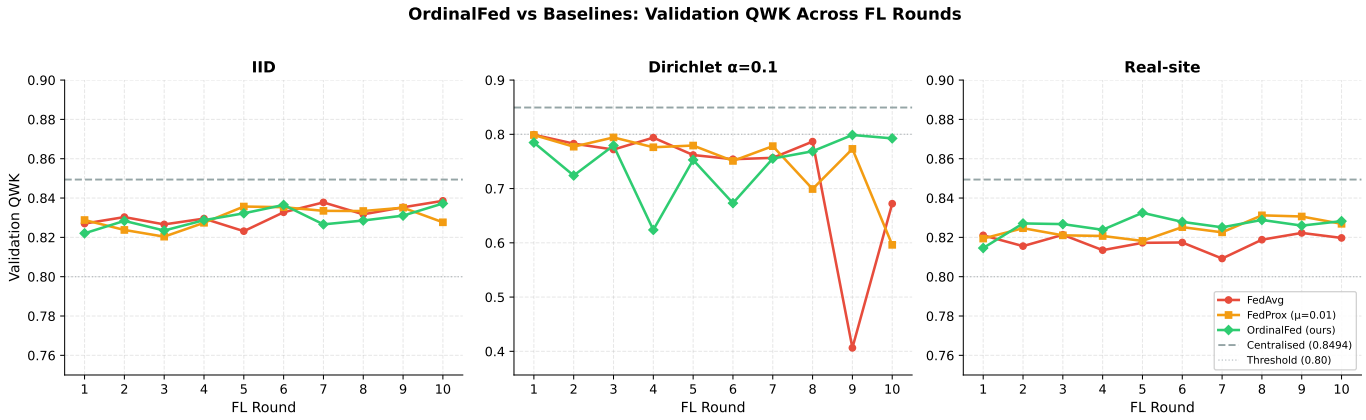


Fig. 2: Validation QWK trajectories across 10 communication rounds for IID, Dirichlet ($\alpha = 0.1$), and real-site partitions. While FedAvg and FedProx suffer catastrophic collapse under extreme skew ($\alpha = 0.1$), OrdinalFed (green curve) maintains robust stability (QWK ≥ 0.6238) while achieving the highest peak performance in the realistic real-site partition.

F. Limitations

Due to severe natural dataset scarcity in the DDR benchmark (Grade 1 represents only 5.0% of the training set), Grade 1 sensitivity remains below 0.15 across all tested methods. Furthermore, the 50-image server set $\mathcal{V}$ introduces minor QWK estimation noise in final rounds when $\beta(t) \to 0$. Setting a non-zero β lower bound and expanding $\mathcal{V}$ via generative augmentation are promising directions for future work.

VI. Conclusion

OrdinalFed, an ordinal-aware federated learning framework tailored for multi-grade Diabetic Retinopathy screening across heterogeneous hospital networks, has been presented. By integrating a distance-penalising OrdinalLoss with QWK-weighted adaptive quality aggregation, OrdinalFed decisively eliminates the catastrophic stability collapses observed in standard FedAvg and FedProx under extreme statistical skew ($\alpha = 0.1$). Specifically, it reduces peak-to-trough fluctuation from 0.3929 down to 0.1751, ensuring diagnostic consistency. In realistic multi-tier hospital partitions, OrdinalFed achieves a peak QWK of 0.8325, narrowing the gap to the centralised ceiling to just 1.69 pp while strictly preserving patient privacy and adding zero communication overhead. Future work will explore formal differential privacy bounds, cross-dataset validation on APTOS 2019, and federated Grad-CAM interpretability to facilitate clinical deployment.

References

[1] International Diabetes Federation, *IDF Diabetes Atlas*, 10th ed. Brussels, Belgium: IDF, 2021. [Online]. Available: https://diabetesatlas.org (accessed Oct. 7, 2026).

[2] T. Li, Y. Gao, K. Wang, S. Guo, H. Liu, and H. Kang, "Diagnostic assessment of deep learning algorithms for diabetic retinopathy screening," *Inf. Sci.*, vol. 501, pp. 511–522, Oct. 2019, doi: 10.1016/j.ins.2019.06.011.

[3] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Agüera y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. 20th Int. Conf. Artif. Intell. Statist. (AISTATS)*, vol. 54, 2017, pp. 1273–1282, doi: 10.48550/arXiv.1602.05629.

[4] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in *Proc. Mach. Learn. Syst. (MLSys)*, vol. 2, 2020, pp. 429–450, doi: 10.48550/arXiv.1812.06127.

[5] S. Toledo-Cortés, D. H. Useche, H. Müller, and F. A. González, "Grading diabetic retinopathy and prostate cancer diagnostic images with deep quantum ordinal regression," *Comput. Biol. Med.*, vol. 145, Art. no. 105472, Jun. 2022, doi: 10.1016/j.compbiomed.2022.105472.

[6] G. Polat, Ü. M. Çağlar, and A. Temizel, "Class distance weighted cross entropy loss for classification of disease severity," 2024, *arXiv:2412.01246*, doi: 10.48550/arXiv.2412.01246.

[7] D. Bhulakshmi and D. S. Rajput, "FedDL: Personalized federated deep learning for enhanced detection and classification of diabetic retinopathy," *PeerJ Comput. Sci.*, vol. 10, Art. no. e2508, Dec. 2024, doi: 10.7717/peerj-cs.2508.

[8] G. Ciaramella *et al.*, "A privacy-preserving and explainable method for diabetic retinopathy classification and localization," *IEEE Access*, vol. 14, pp. 49830–49850, 2026, doi: 10.1109/ACCESS.2026.3678201.

[9] I. Bejenar, L. Ferariu, C. Pascal, and C.-F. Caruntu, "Aggregation methods based on quality model assessment for federated learning applications: Overview and comparative analysis," *Mathematics*, vol. 11, no. 22, Art. no. 4610, Nov. 2023, doi: 10.3390/math11224610.

[10] Z. Li, T. Lin, X. Shang, and C. Wu, "Revisiting weighted aggregation in federated learning with neural networks," in *Proc. 40th Int. Conf. Mach. Learn. (ICML)*, vol. 202, 2023, pp. 19767–19788, doi: 10.48550/arXiv.2302.10911.

[11] X. Li, Y. Gao, Y. Deng, and X. Jiang, "Federated learning with adaptive aggregation weights for non-IID data in edge networks," *IEEE Trans. Cogn. Commun. Netw.*, vol. 11, no. 5, pp. 3425–3439, Oct. 2025, doi: 10.1109/TCCN.2025.3534248.

[12] A. Mora, A. Bujari, and P. Bellavista, "Enhancing generalization in federated learning with heterogeneous data: A comparative literature review," *Future Gener. Comput. Syst.*, vol. 157, pp. 1–15, Aug. 2024, doi: 10.1016/j.future.2024.03.027.